
Problem Set 3

Instructions: Answer the following questions in their entirety in a separate document. Submit your completed problem set as a PDF document via email to sean.corcoran@vanderbilt.edu. Use your last name and problem set number as the filename (e.g., *Ginsburg Problem Set 3.pdf*). Working together is encouraged, but it is expected that all submitted work be that of the individual student.

1. **(16 points—2 each)** In a population of students, the number of absences during the school year ranges from 0 to 8. The probabilities of a randomly drawn student from this population having 0, 1, 2, ..., 8 absences are shown in the table below. Define the event A as the student being absent *fewer than* 4 days, and the event B as the student being absent *at least* 2 days.

# of Days	0	1	2	3	4	5	6	7	8
Probability	0.2	0.14	0.25	0.11	0.1	0.09	0.05	0.03	0.03

- (a) What is the probability of event A ?
 - (b) What is the probability of event B ?
 - (c) What is the probability of $\sim A$?
 - (d) Are events A and B mutually exclusive? Explain why or why not.
 - (e) What is the probability of $A \cap B$?
 - (f) What is the probability of $A \cup B$?
 - (g) Show using values from the table that $P((A \cap B) \cup (\sim A \cap B)) = P(B)$.
 - (h) Show using values from the table that $P(A \cup (\sim A \cap B)) = P(A \cup B)$.
2. **(6 points—3 each)** Using the probability distribution in Question 1, find the following (and show your work):
- (a) $E(\# \text{ of absences})$
 - (b) $Var(\# \text{ of absences})$ and $SD(\# \text{ of absences})$

3. (8 points—2 each) Shown below is a 2 x 2 table that reports the fraction of the population in each cell:

		Education level		
		HS	<HS	Totals
Current smoker:	NO	0.614	0.130	0.744
	YES	0.194	0.062	0.256
Totals		0.808	0.192	1.000

- For a randomly drawn person, what is $P(\text{smoker})$?
 - For a randomly drawn person, what is $P(\text{smoker} \mid <\text{HS diploma})$?
 - For a randomly drawn person, what is $P(\text{smoker} \mid \text{HS diploma+})$?
 - Are education and smoking status independent? Why or why not?
4. (5 points) After the attacks of September 11, 2001, the TSA implemented a program called SPOT (Screening of Passengers by Observation Techniques) in which passengers were flagged for suspicious behavior and given additional searching or screening. Suppose that:
- There are 2 billion + 100 passenger trips per year (2,000,000,100).
 - 100 of these passengers are terrorists.
 - Nearly all (99%) terrorists exhibit the kinds of behaviors that were flagged.
 - Some non-terrorists exhibit these suspicious behaviors, but it is rare (1%).

The SPOT test has low false negative and false positive rates, suggesting it is an effective way to catch would-be terrorists. Use Bayes' Theorem to calculate the probability that a flagged passenger is, in fact, a terrorist.

5. (4 points) Assume the random variable z has a standard normal distribution. Use Stata, an online calculator, or a textbook table to answer the following:
- The probability is 0.75 that z is less than what number?
 - The probability is 0.30 that z is less than what number?
 - The probability is 0.10 that z is greater than what number?
 - The probability is 0.60 that z is greater than what number?

6. **(6 points)** Applicants to Local U. have SAT composite scores that follow a normal distribution with a mean of 1100 and a standard deviation of 160.
- (a) What is the probability that a Local U. applicant will have a composite SAT score of 1300 or higher?
 - (b) What is the probability that a Local U. applicant will have a composite score above 900 but below 1300?
 - (c) Showing strikingly bad judgment, Local U. wishes to offer a tuition-free scholarship to applicants who score in the top 2% of applicants on the SAT. Above what score will they offer this scholarship?

In each case, show or explain how you obtained your result.

7. **(6 points—3 each)** Suppose the probability that a teenage driver gets into an accident during a one-year period is 0.12, and assume the probability of getting into an accident is independent across drivers.
- (a) A particular family has 5 teenage drivers. What is the probability that *at least one driver* in this family will have an accident over the coming year? Show or explain how you obtained your answer.
 - (b) Now consider the population of families with 5 teenage drivers, and define X as the number of accidents that occurred among their 5 drivers. For these families in a typical year, what is $E(X)$ and $sd(X)$? Show or explain how you obtained your answer.
8. **(9 points—3 each)** Simulated draws from probability distributions in Stata. Create an empty dataset with 1,000 (blank) observations
- (a) Create a variable called *score1* that consists of random draws from the normal distribution with mean 25 and standard deviation 5. After doing so, get the mean and standard deviation of *score1* and produce a histogram for it.
 - (b) Create a variable called *score2* that consists of random draws from the uniform distribution between 10 and 170. After doing so, get the mean and standard deviation of *score2* and produce a histogram for it.
 - (c) Challenge: create a variable called *absences* that consists of random draws from the discrete probability distribution given in part 1. Hint: how can you create a variable that is equal to 0 20% of the time, equal to 1 14% of the time, and so on? After doing so, get the mean and standard deviation of *absences* and produce a histogram for it. (This variable is discrete so you may want to use a bar graph or histogram with the discrete option).